

IRIS CLASSIFICATION PIPELINE

Multi-Class One-vs-Rest (OvR) Logistic Regression Architecture

Machine Learning Pipeline Diagram

1. Dataset Ingestion

IRIS DATASET

150 Samples • 4 Feature Dimensions (Sepal Length/Width, Petal Length/Width) • 3 Target Classes

2. Data Preprocessing & Validation

StandardScaler

Feature Scaling ($\mu=0, \sigma=1$)

80 / 20 Train Split

Stratified Train-Test Split

Cross-Validation

K-Fold Model Validation

3. Multi-Class Target Mapping (3 Classes)

Class 0: Setosa | Class 1: Versicolor | Class 2: Virginica

ONE-VS-REST (OvR) DECOMPOSITION STRATEGY

Model 1

Setosa vs Rest

$y_1 = (Class == 0)$

Model 2

Versicolor vs Rest

$y_2 = (Class == 1)$

Model 3

Virginica vs Rest

$y_3 = (Class == 2)$

3 PARALLEL IMPLEMENTATION STRATEGIES

1. Scikit-Learn

Standard

`LogisticRegression(multi_class='ovr')`
optimized via L-BFGS.

2. Custom Perceptron

Gradient descent inspired custom model using
weighted linear decision boundary.

3. Custom Sigmoid

From-scratch implementation using $1 / (1 + e^{-z})$ & Binary Cross-Entropy Loss.

Probability / Score Computation

Calculate $P(y = k | X)$ for each model $k \in \{1, 2, 3\}$

Final Prediction (Argmax)

$\hat{y} = \operatorname{argmax}_k (P_k)$

Performance Comparison & Experimental Report

Evaluation via Accuracy, Precision, Recall, F1-Score & Confusion Matrix analysis across all three implementation strategies.